BlastSmart: A Mobile Application for Blast Design and Fragmentation Analysis

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1 Abstract

BlastSmart is a mobile application developed to simplify, improve, and increase the accuracy of blast calculations used in surface mining. The application consists of a Surface Blasting Design Calculator, which calculates the burden, stemming length, and actual powder factor, and a Fragmentation Calculator, which estimates the mean fragmentation size using data from the Surface Blasting Design Calculator. By reducing the need for manual calculations, the app helps mining engineers make better decisions while improving calculation accuracy, safety, and productivity.

2 Background

The mining industry is using more technology to improve safety, accuracy, and productivity. Blast design is one area that benefits from technology because engineers must perform several calculations before blasting. These calculations directly affect rock fragmentation, productivity, blasting costs, and worker safety.

Many engineers still perform blast calculations using pen and paper, which is time-consuming and increases the risk of human error. There is therefore a need for a simple and reliable digital solution that streamlines the calculation process while reducing human errors. BlastSmart is made to address this need by replacing manual calculations with a faster and more efficient digital approach.

3 Literature review

Surface blasting is widely used in the mining industry to break hard rock into smaller pieces for easier excavation and transportation. A successful blast depends on careful planning and accurate calculations. Engineers must determine values such as burden, stemming length, powder factor, and fragmentation size to ensure that the blast is safe, effective, and economical.

According to Konya and Walter (1990), accurate blast calculations are important because they affect the success of a blast. Poor calculations can lead to oversized rocks, higher costs, damage to equipment, and unsafe working conditions. This is why mining engineers must carefully design every blast before it takes place.

In recent years, technology has changed the way blasting calculations are performed. Instead of relying only on manual calculations, many engineers use digital tools and software to improve speed and accuracy. Examples of these tools are JKSimBlast, Blast Management System (BMC), BlastIQ. These tools help reduce human error and make it easier to plan blasting operations. However, many existing programs are expensive, require specialised training, or are only available on desktop computers.

The BlastSmart App is developed to provide a simple mobile solution for blast calculations. It allows users to enter blasting information and automatically calculates important design values. By combining these features into one mobile application, BlastSmart aims to improve blast planning, reduce calculation errors, and help engineers make faster and more informed decisions. The app also supports the mining industry's move towards using modern digital technology to improve blasting operations

4 Design Process

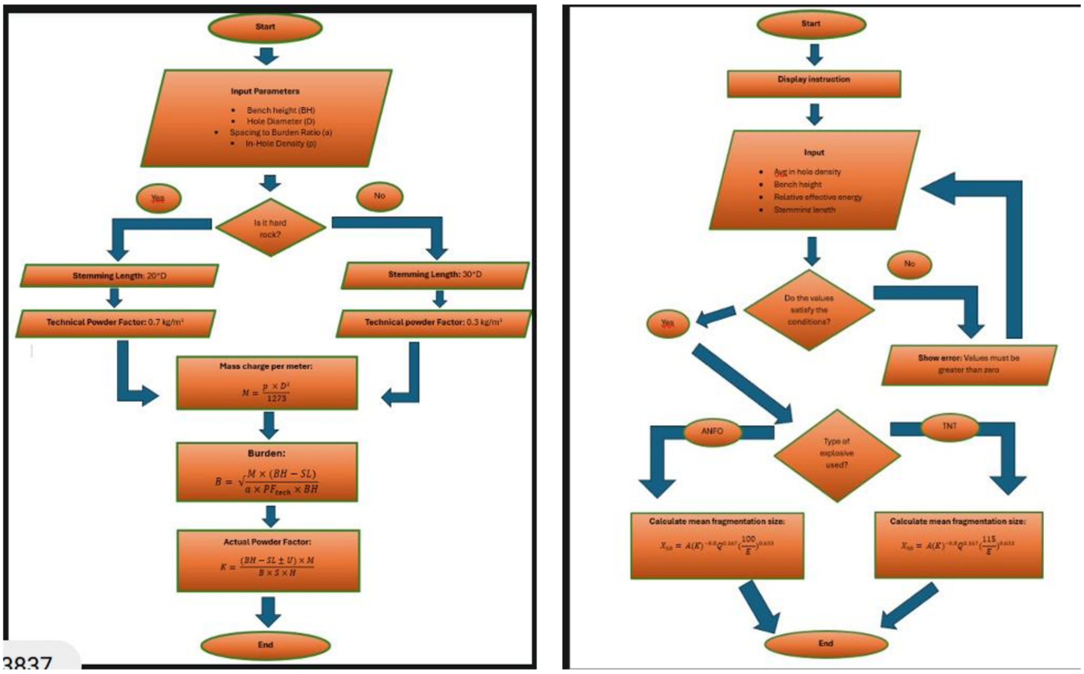

Figure 1: Flow chart for Surface Blasting Design Calculator and Fragmentation size .

5 Results and Discussions

The BlastSmart App the (Surface Blasting Design Calculator and the Fragmentation Calculator). Both calculators produced accurate results based on standard engineering equations after the user entered the required input values. The testing showed that the application performed calculations quickly and consistently, reducing the time and effort required compared to manual calculations.

This app provides a simple, portable and user friendly solution for blast design and fragmentation analysis. It helps reduce calculation errors, improves blast planning. However, the accuracy of the results depends on the accuracy of the input data entered by the user. Especially in the inputting of units, we use the metric system.

6 Conclusion

The BlastSmart mobile application was successfully developed to simplify blast design calculations for mining engineers. The application combines the Surface Blasting Design Calculator and the Fragmentation Calculator into a single mobile platform, allowing users to perform important blasting calculations quickly and accurately.

The project demonstrates that digital technology can reduce the time required for manual calculations, minimise human error, and improve decision-making during blast planning. By providing accurate calculations for burden, stemming length, actual powder factor, and mean fragmentation size, BlastSmart mobile application contributes to safer, more efficient, and more productive mining operations.

Although the application performs well, its accuracy depends on the correctness of the data entered by the user. Overall, BlastSmart provides a strong foundation for the digital transformation of blasting operations and has the potential to become a valuable tool for mining engineers.

7 Future Works

Future versions of BlastSmart can be expanded to provide a complete blasting management system for the mining industry. Additional features may include blast pattern design with graphical visualisation, vibration and air-blast prediction, explosive quantity optimisation, and automatic blast cost estimation, as these are important components of modern blast planning systems (Konya and Walter, 1990; ISEE, 2022).

The application can also be integrated with cloud storage to allow engineers to save, retrieve, and share blast designs across multiple devices. GPS integration could be added to record blast locations, while artificial intelligence and machine learning techniques could be incorporated to recommend optimal blast designs based on previous blasting data. These technologies are becoming increasingly important in digital mining and smart blasting systems (Orica Mining Services, 2023; Hexagon Mining, 2024).

Future development should also include support for underground blasting calculations, additional international blasting standards, and real-time data collection from mining operations. These improvements would make BlastSmart a more comprehensive, intelligent, and industry-ready mobile application for blast planning and analysis (Jimeno et al., 1995; Olofsson, 1990).

8 References

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